

Digital Talkbox:

a MATLAB physical model of the vocal tract

Try the talkbox here: <https://github.com/haleyjanvrin/TalkBox>

Haley Janvrin, April 2025

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Motivation

For this project, I wanted to use a physical modelling approach to make a MATLAB simulation of the vocal tract. The talkbox uses a tube to feed an instrument's output, usually a piano or guitar, into the user's mouth so that they can shape the sound with their mouth, lips, and tongue to mimic vocal sounds. My goal was to replicate this effect in MATLAB and make a program that filters an input audio file based on variable physical parameters dictating mouth shape, as if the sound were passing through the vocal tract.

Physical approaches to vocal tract modelling have been well established since 1962, when Kelly and Lochbaum published the first digitized vocal tract [1]. Synthetic singing was originally based on passing a glottal pulse through these physical models; however, as the performance of alternative spectral models like the vocoder improved and were able to achieve more realistic speech, physical modelling decreased in popularity. With this project, I was more interested in capturing the impact of changing the shape of the mouth than accurately mimicking the human voice, which is what led me to take a physical modelling approach.

Kelly-Lochbaum Acoustic Tube

There are many existing vocal tract models, most of which are based on the original Kelly and Lochbaum acoustic tube model. This approach models the vocal tract as a series of cylinders with varying radii [2]. By changing the size of the cylinders, Kelly and Lochbaum were able to replicate how a person constricts and dilates

sections of the vocal tract to make different sounds and say words. They programmed an IBM 704 computing machine to sing “Daisy Bell (Bicycle Built for Two)” as the first documented instance of physical-based speech synthesis, which would later inspire the computer song in HAL-9000's death scene in hit movie *2001: A Space Odyssey* [1].

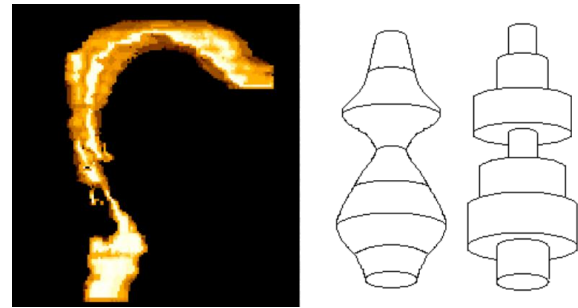


Figure 1: Profile of a male vocal tract making the [ae] vowel sound, composed from MRI scans (left) [4]. 7-section Kelly-Lochbaum acoustic tube model of the vocal tract, both smoothed and sampled (right) [4].

When a wave travels from the glottis to the lips, Kelly and Lochbaum assume lossless scattering at each junction between cylindrical sections [2]. Due to the area discontinuities at these junctions, the wave is partially reflected backwards towards the glottis and partially propagated forwards towards the lips [3]. Therefore, in each section we can define the total pressure P made up of two components: the forward wave $P+$ and the backward wave $P-$.

Each tube section has a characteristic impedance or resistance to airflow R_k , based on its cross-sectional area A_i [3].

$$R_k = \frac{\rho c}{A_k} \quad [1]$$

The reflection coefficient r at the intersection of two tubes k and $k+1$ is defined by the characteristic impedances of the tubes.

$$r_{forward} = \frac{R_{k+1} - R_k}{R_{k+1} + R_k} \quad [2]$$

Using the relationship defined in equation 1 assuming no losses, this simplifies to a ratio of the cross-sectional areas.

$$r_{forward} = \frac{A_k - A_{k+1}}{A_k + A_{k+1}} \quad [3]$$

At the junction between cylindrical sections, the forward wave P^+ is partially reflected backwards and partially propagated according to the respective impedances. This reflected fraction, rP^+ , joins the backward wave. Therefore, since P^+ is continuous at the interface the propagated fraction must be $(1+r)P^+$ to satisfy continuity.

$$P^+ = (1 + r)P^+ - rP^+ = P^+ \quad [4]$$

The propagation of the backward wave is calculated in a similar manner. However, since it is moving in the opposite direction, the reflection coefficient is calculated by swapping the impedances in equation 2.

$$r_{reverse} = \frac{R_k - R_{k+1}}{R_k + R_{k+1}} = -r_{forward} \quad [5]$$

Therefore, using r to denote the reflection coefficient going forward, the reflected fraction of P^- is $(-r)P^-$ and joins the forward wave. The propagated fraction is thus $(1-r)P^-$ to maintain continuity. This relationship between the reflected and propagated portions of the forward and backward waves is illustrated in Fig. 2.

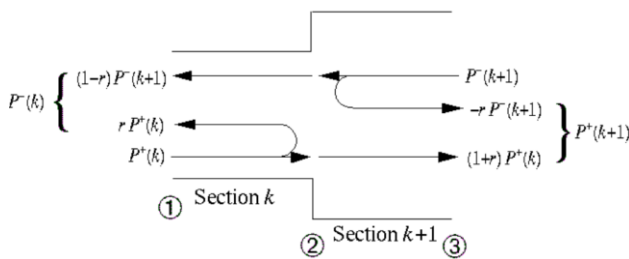


Figure 2: Reflected and propagated fractions at the junction between tube sections [3].

Project Design

The Kelly-Lochbaum acoustic tube model for the vocal tract can be translated into a type of digital waveguide model [1]. This implements the sampled d'Alembert's traveling-wave solution in each section of the piece-wise cylindrical tube model, as outlined below in Fig. 3 [1]. R_i represents the characteristic impedance of each tube section, and k_i is the forward reflection coefficient of each junction. Delay lines are implemented to represent the wave's travelling through each section.

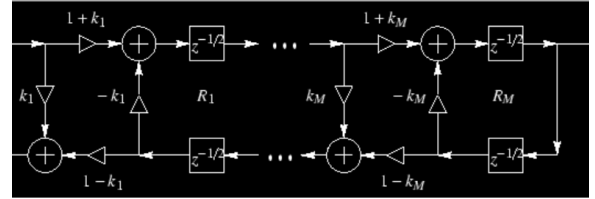


Figure 3: Digital waveguide model of the Kelly-Lochbaum vocal tract [1].

The upper track in the model represents the forward traveling wave, and the lower track the backward wave. For synthetic singing applications, the model in Fig. 3 takes a glottal pulse input in the top left corner to the forward track, and the speech output is from the top right corner from the forward track. The output from the bottom left corner on the backward track is an unused all-pass filtered output [1].

To make my vocal tract model, I implemented this digital waveguide model in MATLAB. Instead of a glottal pulse, it takes an audio file as input to the forward wave. The reflection coefficients are calculated at each junction between cylinder sections, and the length of the delay lines is equal to the time in samples it takes sound to pass through each section.

To set the cross-sectional area of each tube section, I used data from Baer et al. They composed estimations of vocal tract dimensions for four different vowel sounds from MRI scans, reporting on the cross sectional area at 19 regular intervals from the larynx.

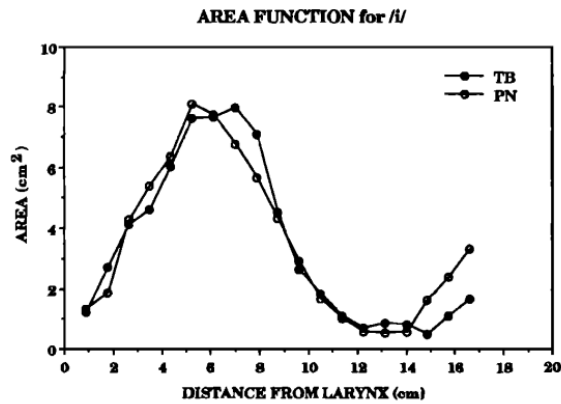


Figure 4: Cross-sectional area of the vocal tract for the /i/ vowel sound [5].

These area estimations create a 19-segment Kelly-Lochbaum vocal tract. The parameters of these four vowel sounds (/a/, /æ/, /i/, /u/) are included as options in the digital talkbox. The user selects one of the preset vowel vocal tract shapes, and the input audio is filtered accordingly.

Challenges

My largest challenge with this project was by far the implementation. The digital waveguide model is relatively complicated to implement in MATLAB as there are a lot of feedback and feedforward components. Although I believe my implementation is correct and despite my best attempts at debugging, I am still having a problem where the filtered output signal grows exponentially and very quickly to infinity.

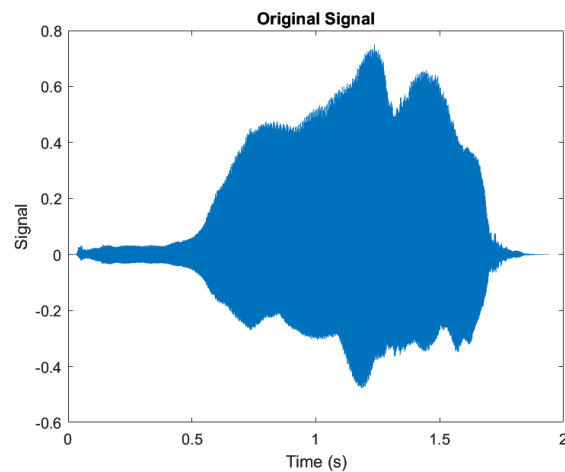


Figure 5: The original sample audio input signal.

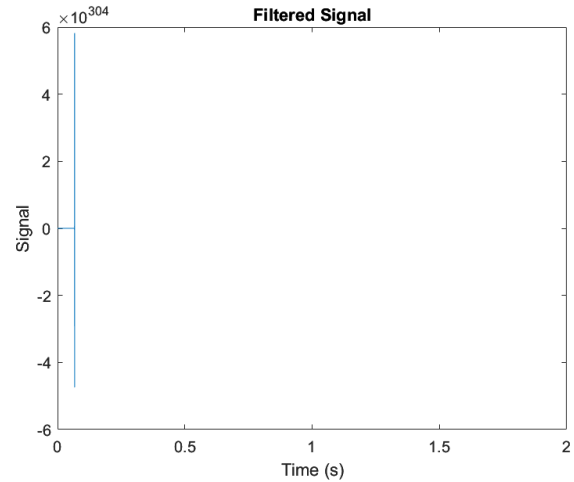


Figure 6: The attempted filtered output signal using the /i/ vowel shape.

When this problem initially persisted, I thought the cross-sectional area values I was using might be the issue since they were based on experimental data. I tried instead using vocal tract radius values for a 10-segment acoustic tube model from Perry Cook's thesis on synthetic singing [3]. He produced a similar although much more complicated version of my attempted vocal tract model, using the FAST tracker on spoken phonation samples to obtain vocal tract shapes for each vowel sound. However, although the simpler 9-segment model reached infinity slightly slower than the original 19-segment model, the model's response was still clearly inaccurate.

The issue of a filter response inexplicably reaching infinity even when I believe my implementation to be correct is something I encountered many times throughout this class, in both the assignments and midterms. It may be an error in the way I code in MATLAB, and is a problem that I am still actively trying to understand and solve.

References

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